

CUSTOM ALERTS MINI REPORT

1. Executive Summary

This report outlines the technical procedures and specific logic we used to implement custom detection using Microsoft Sentinel. The aim is to translate historical Indicators of Compromise (IoCs) and observed attacker behaviors from recent security incidents into proactive, automated analytics rules.

2. Infrastructure Prerequisites

Before deploying the custom alerts, certain things were put in order in Azure. First, we created a dedicated Resource Group (**CyblackTeam8**) in the East US Region.

☐	Name ↑		Subscription	Location
☐	 CyblackTeam8	...	Azure subscription 1	East US

Next, a Log Analytics Workspace was created for Microsoft Sentinel in the same region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Logs

Resource visualizer

Settings

Classic

Azure Monitor will enforce TLS versions 1.2 and above starting March 1, 2026. To avoid data loss and service disruption please **upgrade to AMA** and ensure your OS supports TLS 1.2 or higher. [Learn More About TLS](#), [Learn More About Agents](#).

Essentials

Resource group [\(move\)](#)
cyblackteam8

Status
Active

Location
East US

Subscription [\(move\)](#)
[Azure subscription 1](#)

Workspace Name
LogAnalyticsCyblackTeam8

Workspace ID
[REDACTED]

Pricing tier
Pay-as-you-go

Access control mode
Use resource or workspace permissions

JSON View

Then, we added the Workspace to Microsoft Sentinel

Home

Microsoft Sentinel

List supported metrics for these resources

Filter resources by region

Review service health across resources

Default Directory [REDACTED]

Create

Manage view

Refresh

Export to CSV

Open query

View incidents

Add to service group

Group by none

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

☐	Name ↑	Resource Group	Location	Subscription	Directory	
☐	 LogAnalyticsCyblackTeam8	...	cyblackteam8	East US	Azure subscription 1	Default Directory

Next, we established necessary Data Connectors in Microsoft Sentinel.

Microsoft Sentinel | Data connectors ...

Selected workspace: 'loganalyticscyblackteam8'

Search Refresh Guides & Feedback

General
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings

7 Connectors 7 Connected More content at Content Hub

Search by name or provider Providers : Microsoft Data Types : All

Status : Connected (7)

	Microsoft 365 Insider Risk Management (Preview) Microsoft	...
	Microsoft Defender for Cloud Apps Microsoft	...
	Microsoft Defender for Endpoint Microsoft	...
	Microsoft Defender for Identity Microsoft	...
	Microsoft Defender for Office 365 (Preview) Microsoft	...
	Microsoft Defender XDR Microsoft	...

Microsoft Defender XDR: This is the primary umbrella connector. It bidirectionally synchronizes comprehensive incidents, alerts, and raw advanced hunting data from across the entire Microsoft Defender security stack into Sentinel.

Microsoft Defender for Endpoint: The aim of this connector is to Ingest endpoint detection and response (EDR) telemetry. This gives Sentinel deep insights into device-level activities, such as malicious process executions, registry modifications, and suspicious network connections.

Microsoft Defender for Identity: This connector streams alerts and behavioral data related to compromised credentials, lateral movement, and domain-level attacks across your on-premises Active Directory and Entra ID environments.

Microsoft Defender for Cloud Apps: This pulls in discovery logs and alerts regarding anomalous activity within connected SaaS cloud applications. This is crucial for monitoring shadow IT, unauthorized data access, and abnormal download patterns.

Microsoft Defender for Office 365 (Preview): Ingests telemetry and alerts related to email and collaboration threats. This provides data on phishing campaigns, malicious attachments, and weaponized links sent via Exchange Online, SharePoint, OneDrive, and Teams.

Microsoft 365 Insider Risk Management (Preview): Streams alerts triggered by your M365 compliance policies. This is specifically designed to detect internal threats, such as intellectual property theft, data leakage, or unauthorized data sharing by employees.

3. Custom Analytics Rules Engineering

The following detection rules utilize the Kusto Query Language (KQL) to scan ingested logs for specific threats identified in the incident reports.

3.1 Alert: Suspicious Foreign Authentication Attempts

Objective: Detect brute-force or successful authentication attempts originating from known malicious infrastructure observed during the Erwin Smith account compromise.

- **Intelligence Source:** Identified attacker IPs originating from Bulgaria, China, and Thailand.
- **MITRE ATT&CK:** T1110 (Brute Force).
- **Data Source:** SigninLogs

The screenshot shows the 'Analytics rule wizard - Create a new Scheduled rule' interface in Microsoft Sentinel. The breadcrumb navigation at the top reads: 'Connectors > Data connectors > Analytics > Analytics rule wizard'. On the left, a vertical sidebar contains five steps: 'General' (selected with a blue dot), 'Set rule logic', 'Incident settings', 'Automated response', and 'Review + create'. The main panel on the right is titled 'Create an analytics rule that will run on your data to detect threats.' and 'Analytics rule details'. It contains the following fields: 'Name' with the value 'Detect Suspicious Foreign Sign-ins (Bulgaria, China, Thailand)'; 'Description' with the value 'Detects authentication attempts from the specific foreign IP addresses'; 'Severity' set to 'High' (indicated by three red squares); 'MITRE ATT&CK' set to '2 Selected'; and a 'Status' section. At the bottom, there is a blue button labeled 'Next : Set rule logic >' and a grey 'Cancel' button.

Then the columns generated in KQL query was linked to Sentinel's built-in entity types via Alert Enhancement.

Validation passed.



Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create**

Analytics rule details

Name Detect Suspicious Foreign Sign-ins (Bulgaria, China, Thailand)

Description Detects authentication attempts from the specific foreign IP addresses

MITRE ATT&CK [Credential Access \(1\)](#)

Severity ■ ■ ■ High

Status ☒ Enabled

Analytics rule settings

Rule query SigninLogs | where IPAddress in ("79.124.60.6", "150.158.77.170", "124.120.140.202") | project TimeGenerated, UserPrincipalName, IPAddress, Location, AppDisplayName, ResultType

Rule frequency Run query every **5 minutes**

Rule period Last **5 minutes** data

Rule start time Automatic

< Previous

Save

Cancel

Full KQL code snippet

SigninLogs

```
| where IPAddress in ("79.124.60.6", "150.158.77.170", "124.120.140.202")  
| extend City = tostring(LocationDetails.city), Country =  
tostring(LocationDetails.countryOrRegion)  
| project TimeGenerated, UserPrincipalName, ResultType, IPAddress, City,  
Country, AppDisplayName  
| sort by TimeGenerated desc
```

3.2 Alert: Malware Command & Control (C2) Communication

Objective: Identify endpoint traffic communicating with the AsyncRAT and Raccoon Stealer infrastructure.

- **Intelligence Source:** Primary and secondary C2 IP addresses utilized for credential exfiltration.
- **MITRE ATT&CK:** T1071 (Application Layer Protocol).
- **Data Source:** CommonSecurityLog

Connectors > Data connectors > Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

- ☒ General
- ☐ Set rule logic
- ☐ Incident settings
- ☐ Automated response
- ☐ Review + create

Analytics rule details

Name *

Description

Severity

High

MITRE ATT&CK

Command And Control

Status

☒ Enabled

Next : Set rule logic >

Cancel

Connectors > Data connectors > Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic**
- Incident settings
- Automated response
- Review + create

Define the logic for your new analytics rule.

Rule query
Any time details set here will be within the scope defined below in the Query scheduling fields.

```
CommonSecurityLog
| where DestinationIP in ("128.24.231.64", "128.24.231.65")
| project TimeGenerated, SourceIP, DestinationIP, DestinationPort, RequestURL, DeviceAction
```

[View query results >](#)

Alert enhancement

- > Entity mapping
- > Custom details
- > Alert details

Query scheduling

< Previous **Next : Incident settings >** Cancel

Connectors > Data connectors > Analytics > Analytics rule wizard

Validation passed.

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic
- Incident settings**
- Automated response
- Review + create

Analytics rule details

Name	Detect Outbound Malware C2 Traffic
Description	Detects outbound network connections to the known Command and Control (C2) IP addresses associated with the recent AsyncRAT and Raccoon Stealer infection.
MITRE ATT&CK	Command And Control
Severity	High
Status	Enabled

Analytics rule settings

Rule query	CommonSecurityLog where DestinationIP in ("128.24.231.64", "128.24.231.65") project TimeGenerated, SourceIP, DestinationIP, DestinationPort, RequestURL, DeviceAction
Rule frequency	Run query every 5 minutes
Rule period	Last 5 minutes data
Rule start time	Automatic

< Previous **Save** Cancel

Full KQL Code snippet

CommonSecurityLog

| where DestinationIP in ("128.24.231.64", "128.24.231.65")

| project TimeGenerated, SourceIP, DestinationIP, DestinationPort, RequestURL, DeviceAction, DeviceName

Incident Response Action: Isolate the originating endpoint (SourceIP) via EDR API integration and generate a High-Severity incident ticket.

3.3 Alert: Exfiltration of Highly Sensitive Data

Objective: Detect unauthorized downloads of critical organizational assets from SharePoint, explicitly mimicking the exfiltration observed in the recent breach.

- **Intelligence Source:** Download of employee financial data and internal security assessments.
- **MITRE ATT&CK:** T1041 (Exfiltration Over Web Services) and T1005 (Collection).
- **Data Source:** OfficeActivity

Connectors > Data connectors > Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

- General**
- Set rule logic
- Incident settings
- Automated response
- Review + create

Create an analytics rule that will run on your data to detect threats.

Analytics rule details

Name *

Description

Severity

High

MITRE ATT&CK

2 Selected

Status

[Next : Set rule logic >](#)[Cancel](#)

Connectors > Data connectors > Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic**
- Incident settings
- Automated response
- Review + create

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
OfficeActivity
| where RecordType == "SharePointFileOperation"
| where Operation == "FileDownloaded"
| where SourceFileName in~ ("updated_payroll.xlsx", "vulnerability_report_June2024")
| project TimeGenerated, UserId, ClientIP, SourceFileName, OfficeObjectId
```

[View query results >](#)

Please wait while we evaluate your query...

Alert enhancement

- > Entity mapping
- > Custom details
- > Alert details

Query scheduling

[< Previous](#)[Next : Incident settings >](#)[Cancel](#)

Connectors > Data connectors > Analytics > Analytics rule wizard

Validation passed.

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Analytics rule details	
Name	Detect Sensitive Payroll or Vulnerability File Download
Description	Triggers when specific high-value internal documents (payroll data or vulnerability reports) are downloaded from SharePoint, indicating potential data exfiltration.
MITRE ATT&CK	- Collection - Exfiltration
Severity	High
Status	Enabled
Analytics rule settings	
Rule query	OfficeActivity where RecordType == "SharePointFileOperation" where Operation == "FileDownloaded" where SourceFileName in ("updated_payroll.xlsx", "vulnerability_report_June2023.docx") project TimeGenerated, UserId, ClientIP, SourceFileName, OfficeObjectid
Rule frequency	Run query every 10 minutes

< Previous Save Cancel

Full KQL Code Snippet

OfficeActivity

```
| where RecordType == "SharePointFileOperation"
| where Operation == "FileDownloaded"
| where SourceFileName in ("updated_payroll.xlsx",
"vulnerability_report_June2023.docx")
| project TimeGenerated, UserId, ClientIP, SourceFileName, Site_Url,
UserAgent
```

3.4 Alert: Malicious Use of Legitimate Microsoft Domains (LotL)

Objective: Detect the specific network anomalies where attackers used legitimate Microsoft telemetry domains to mask data tunneling.

- **Intelligence Source:** Malware traffic directed to specific Windows data endpoints using TLS 1.2.
- **Data Source:** CommonSecurityLog

Analytics rule wizard - Create a new Scheduled rule

General☐ Set rule logic☐ Incident settings☐ Automated response☐ Review + create

Create an analytics rule that will run on your data to detect threats.

Analytics rule details**Name ***

Detect LotL: Malicious Use of Microsoft Domains (login.live / settings-win)

Description

Detects potentially malicious outbound data tunneling and credential exfiltration to legitimate Microsoft domains, as observed in the AsyncRAT/Raccoon Stealer infection.

Severity

High

MITRE ATT&CK

2 Selected

Status

Next : Set rule logic >

Cancel

Analytics rule wizard - Create a new Scheduled rule

☐ General**☒ Set rule logic**☐ Incident settings☐ Automated response☐ Review + create

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
// Define the suspicious domains in a list for cleaner code and easier updating
let TargetDomains = dynamic(["login.live.com", "settings-win.data.microsoft.com", "CommonSecurityLog"])
// 1. Filter early: Check both fields in case your specific firewall/proxy parses
// where DestinationHostName has_any (TargetDomains) or RequestURL has_any (TargetDomains)
// 2. Apply the exfiltration threshold BEFORE projecting to optimize memory usage
```

[View query results >](#)**Alert enhancement**

> Entity mapping

> Custom details

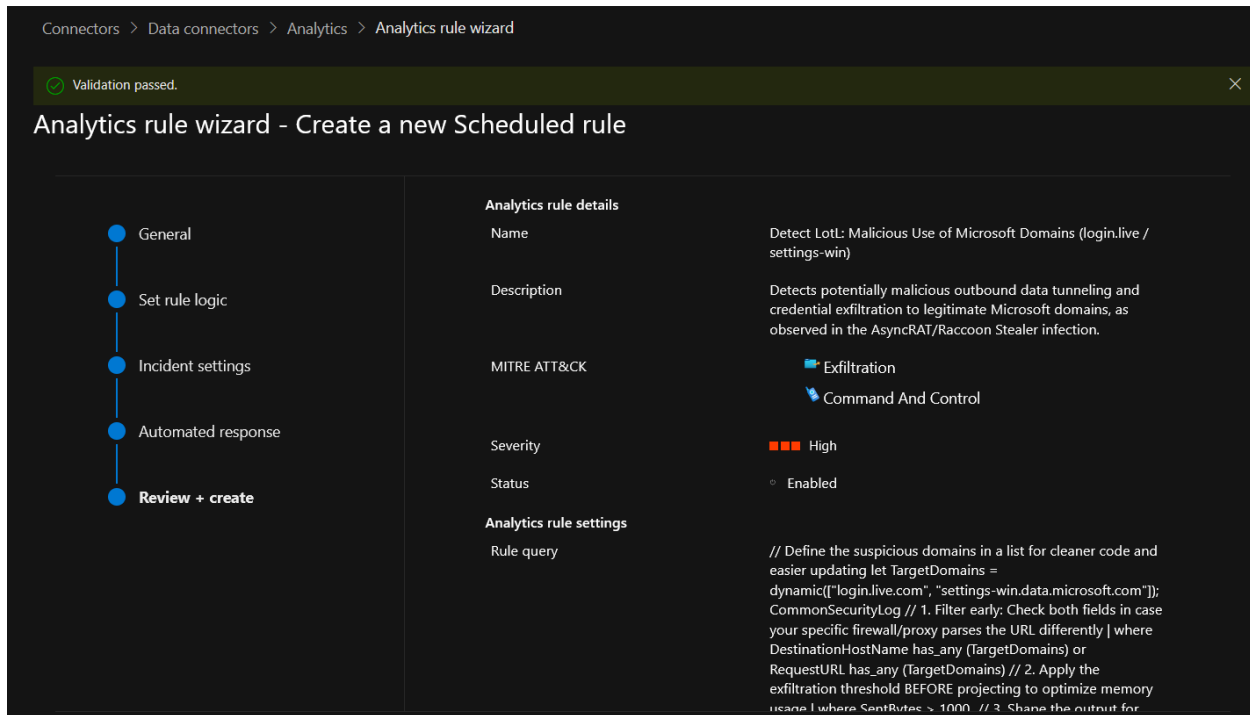
> Alert details

Query scheduling

< Previous

Next : Incident settings >

Cancel



Full KQL Code Snippet:

```
// Define the suspicious domains in a list for cleaner code and easier updating
let TargetDomains = dynamic(["login.live.com",
"settings-win.data.microsoft.com"]);
CommonSecurityLog
// 1. Filter early: Check both fields in case your specific firewall/proxy parses
the URL differently
| where DestinationHostName has_any (TargetDomains) or RequestURL
has_any (TargetDomains)
// 2. Apply the exfiltration threshold BEFORE projecting to optimize memory
usage
| where SentBytes > 1000
// 3. Shape the output for the analyst
| project TimeGenerated, SourceIP, DestinationIP, DestinationPort,
DestinationHostName, RequestURL, SentBytes, DeviceAction
// 4. Sort to surface the most recent activity first
| sort by TimeGenerated desc
```

4. Deployment Methodology

To implement these rules in your new Azure subscription, follow this standard deployment pipeline:

1. **Rule Creation:** Navigate to **Microsoft Sentinel > Analytics > Create > Scheduled query rule**.
2. **Scheduling:** Set the query frequency to run every 5 to 15 minutes, looking back over the same time period to ensure no gaps in coverage.

3. **Entity Mapping:** Within the analytics rule wizard, map entities such as Account (to UserPrincipalName) and IP (to IPAddress or SourceIP). This allows Sentinel to correlate these alerts with other events to build visual investigation graphs.
4. **Tuning Phase:** Run the alerts in "Testing" mode for 48 hours to establish a baseline and filter out any false positives before enabling automated incident creation.

5. Conclusions

The implementation of these custom Microsoft Sentinel analytics rules marks a critical shift for the organization, transitioning our security posture from reactive incident analysis to proactive threat detection. By codifying the specific Indicators of Compromise (IoCs) and adversary behaviors observed during the recent account compromise and malware infections, the Security Operations Center (SOC) is now positioned to automatically detect and intercept these exact attack vectors before they can inflict operational damage.

To maximize the efficacy of these newly deployed alerts and ensure they provide high-fidelity intelligence, the following next steps are recommended:

- **Continuous Rule Tuning:** The engineering team must closely monitor the alerts during their initial 48-hour testing phase. Adjusting thresholds (such as the byte-count limit for data exfiltration) will be essential to minimize false positives and prevent alert fatigue among analysts.
- **SOAR Integration (Playbooks):** Detection is only the first half of the equation. The team should prioritize developing Azure Logic Apps to serve as automated playbooks. This will allow Sentinel to execute immediate containment actions—such as isolating a compromised endpoint via Carbon Black or automatically revoking Entra ID session tokens—the moment an alert fires.
- **Dynamic Threat Intelligence:** While these custom rules are highly effective for known threats, they rely on static IP addresses and file names. The analytics environment should be upgraded to ingest dynamic Threat Intelligence (TI) feeds, allowing Sentinel to cross-reference logs against continuously updated lists of malicious infrastructure.

Ultimately, deploying these targeted detection mechanisms closes the visibility gaps exploited in recent incidents, hardening the network against recurrence and ensuring a more resilient, automated defense strategy moving forward.